

Basic GROQ Inferencing

```
!pip install -q groq
```

```
import os
os.environ["GROQ_API_KEY"] =
"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"
```

```
from groq import Groq
```

```
client = Groq()
```

```
models = [
    "openai/gpt-oss-120b",
    "llama-3.1-8b-instant",
    "moonshotai/kimi-k2-instruct-0905"
]
```

```
prompt = "Explain Generative AI in one short paragraph."
```

```
for model in models:
```

```
    print("\n=====")
    print("Model:", model)
    print("=====")
```

```
    response = client.chat.completions.create(
        model=model,
        messages=[{"role": "user", "content": prompt}],
        temperature=0.7,
        max_tokens=200,
    )
```

```
    print(response.choices[0].message.content)
```

Output

```
=====
Model: openai/gpt-oss-120b
=====
```

Generative AI refers to machine-learning systems—most often deep neural networks such as diffusion models, transformers, or variational autoencoders—that learn the statistical patterns of large datasets (e.g., text, images, audio, code) and can then create new, coherent content that mimics those patterns. By conditioning on prompts or latent

variables, these models can synthesize realistic text, artwork, music, or even functional software, enabling applications from chatbots and virtual assistants to creative design tools and data-augmentation pipelines.

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Model: llama-3.1-8b-instant

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Generative AI refers to a class of artificial intelligence models that can create new, original content, such as text, images, audio, or videos, based on patterns and structures learned from existing data. These models use algorithms like Generative Adversarial Networks (GANs) or Variational Autoencoders (VAEs) to generate new content that is often indistinguishable from human-created work. This technology has numerous applications, including content generation, data augmentation, and creative tasks, and has the potential to revolutionize various industries, from art and media to marketing and education.

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Model: moonshotai/kimi-k2-instruct-0905

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Generative AI is a branch of artificial intelligence that learns patterns from existing data and then creates brand-new content—text, images, audio, code, or other media— that mimics those patterns, effectively producing original output on demand.